Construction and Verification of Long Hairpin RNA Plasmids for Recombinant siRNA Production in E. coli

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ABSTRACT

RNA interference (RNAi) enables targeted gene silencing through siRNAs, which bind to complementary mRNA and suppress its expression. To develop a more efficient siRNA production method, we applied a bacterial system to generate recombinant siRNAs, named pro-siRNAs, from long hairpin RNA constructs. In this project, we designed and cloned hairpin plasmids targeting several cancer-related genes using sequential ligation into the p323 plasmid, followed by transformation and validation in E. coli. Optimization of enzymatic digestion times and enzyme concentrations enhanced cloning efficiency and reproducibility. These results show a reliable and cost-effective approach for producing functional siRNAs in bacteria, building a basis for identifying genes essential for cancer cell survival.

INTRODUCTION

Conventional siRNA approaches require extensive sequence optimization and are costly for large-scale studies. To address this, our lab developed a pro-siRNA system that produces siRNAs directly in bacteria—these pro-siRNAs, processed from double-stranded RNA, offer higher specificity/efficiency without complex design.

Objective

The objective of this research was to establish and optimize a bacterial pro-siRNA cloning system capable of producing long hairpin RNA constructs for efficient siRNA generation, laying groundwork for a cancer-focused RNAi screening platform. Long-term, we aim to identify cancer-dependent genes to inform targeted therapy.

METHODS

Primers were designed to target >200 bp sequences of the gene of interest: forward primers contained SacI/NotI sites, while reverse primers contained XhoI/SalI sites. Using high-fidelity PCR, two DNA fragments were amplified: DNA-1 and DNA-2. The p323 plasmid and DNA-1 were first digested with SacI/XhoI, followed by purification and ligation using T4 DNA ligase. The recombinant plasmids were then transformed into E. coli and plated on antibiotic-containing LB agar for overnight incubation. Next, the verified intermediate plasmids and DNA-2 were digested with NotI/SalI, the ligation, transformation, and colony screening processes were repeated to obtain candidate plasmids. To confirm the final plasmid, triple restriction digestion was performed, and Sanger sequencing was conducted to verify the integrity of the hairpin region.

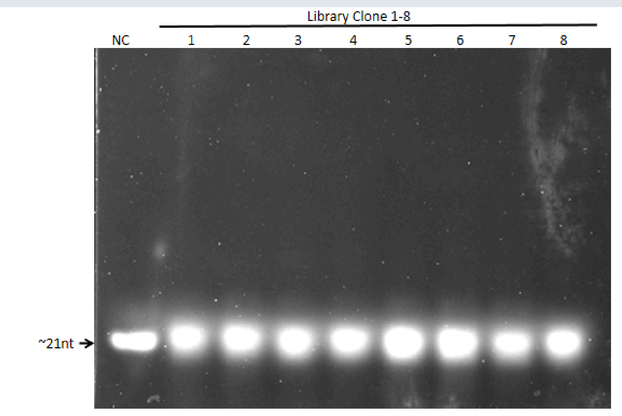


Figure 3. RNA gel electrophoresis showing successful production of siRNA (~21 nt) from library constructs.

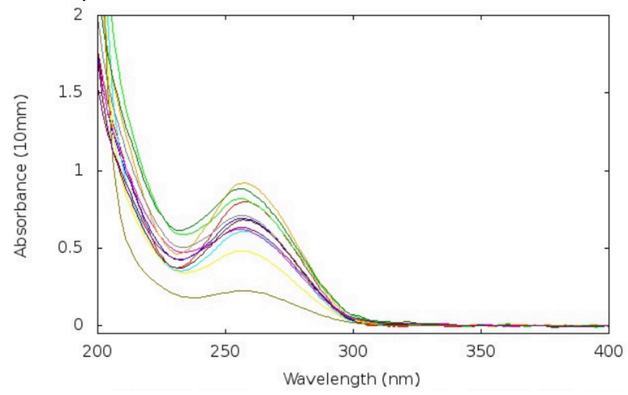


Figure 4. UV absorbance spectra (200 – 400 nm) of siRNA library samples measured after purification. Each curve represents an independent siRNA pool sample, mostly showing consistent A260 peaks near 260 nm.

RESULTS

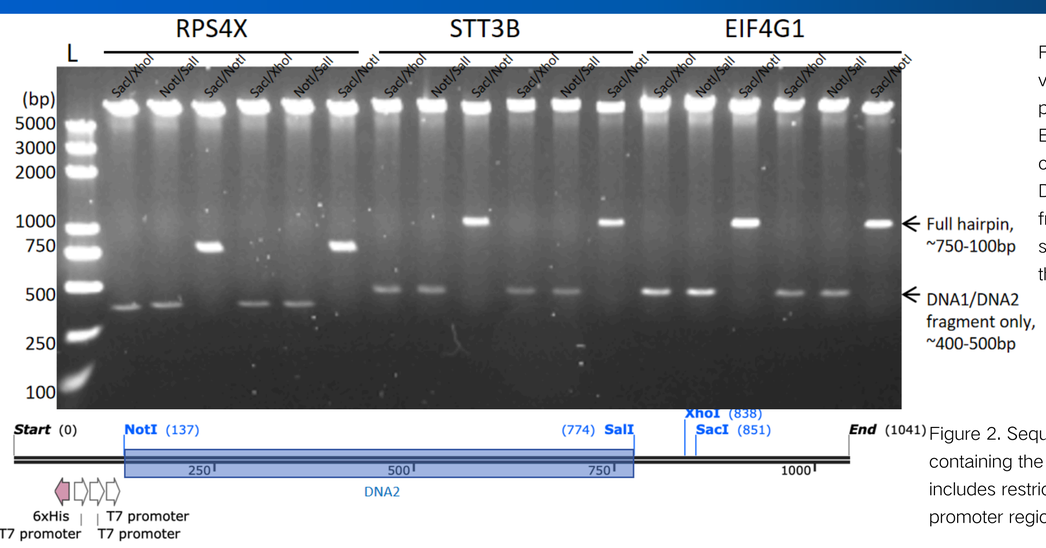


Figure 1. Restriction enzyme validation of long hairpin RNA plasmids for RPS4X, STT3B, and EIF4G1. Bands of ~250 – 500 bp correspond to individual DNA1 or DNA2 fragments, while larger fragments of ~750-100 bp confirm successful ligation of both inserts into the complete hairpin structure.

Figure 2. Sequencing confirmation of the plasmid containing the DNA2 insert. The verified construct includes restriction sites (NotI, SalI, XhoI, SacI) and promoter regions (T7 promoters).

01 Using the pro-siRNA platform, we successfully generated and validated full-length long hairpin RNA constructs targeting 17 cancer-relevant genes. For each gene, a 200 – 689 bp region was amplified via high-fidelity PCR to produce sense and antisense fragments with distinct restriction sites. Sense fragments were ligated into the p323 to form intermediate plasmids, validated by colony PCR, restriction digestion, and sequencing for correct orientation. Antisense fragments were then inserted to yield full hairpin plasmids, with resistant colonies showing digestion patterns matching predicted hairpin sizes. Triple restriction enzyme validation (Figure 1) and Sanger sequencing (Figure 2) further confirmed construct fidelity, with sequencing traces showing high-quality reads and signal attenuation near the SacI junction.

02 The verified plasmid library enables recombinant siRNA production in E. coli. RNA gel electrophoresis showed the successful transcription of siRNA-sized products (~21 nt) from representative library clones, comparable to chemically synthesized siRNA controls (Figure 3). UV absorbance spectra of purified siRNA pools showed consistent A260 peaks near 260 nm across independent preparations, confirmed the RNA purity and concentrations (Figure 4). Together, the constructed plasmids produce intact, correctly sized RNA products, and our purification yields samples are suitable for downstream functional screening of cancer-relevant gene dependencies.

DISCUSSION & FUTURE WORK

In this project, we successfully constructed and validated multiple long hairpin RNA plasmids, demonstrating the feasibility of using the pro-siRNA system for targeted gene silencing. We also refined the cloning protocol: extending restriction digestion time from 1.5 h to 2 h and doubling enzyme/DNA amounts improved digestion efficiency and gel band reproducibility, boosting recombinant siRNA production success in E. coli. Building on this foundation, our next steps will focus on expanding the pro-siRNA library for genome-wide cancer gene screening, applying the system to hepatocellular carcinoma cells to identify functional gene dependencies, and validating top candidate genes using complementary siRNA and CRISPR approaches.

References

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